How-To Library

Informatica

# Address Verification 5.x Best Practices for the United States

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# Abstract

This document tells you how to connect the elements in a United States address to the appropriate address elements in Informatica Address Verification. This document also describes best practices that you can implement when you verify addresses in the United States.

# Supported Versions

- Informatica Address Verification 3.0 and later
- Informatica Address Verification (On-Premises) 5.x

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# Introduction

Informatica Address Verification can validate and enhance addresses in the United States. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for the United States address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to identify the different types of address information in the input data.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

An address in the United States can contain different types of information, including street information and locality information. You must map each item of address information to an appropriate element in Address Verification.

# Process Modes

Informatica Address Verification supports the following process modes for United States addresses:

- BATCH
- CERTIFIED
- COUNTRY RECOGNITION

- FAST COMPLETION
- INTERACTIVE
- PARSE

## Character Set

The reference address databases for the United States store address data in the Latin-1 script. By default, Informatica Address Verification returns the verified United States addresses in the Latin-1 script.

## Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The United States Postal Service recognizes the following structure for a postal address:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: HOUSE NUMBER STREET NAME [SUB-BUILDING INFORMATION]
Line 3: CITY STATE ZIP CODE
Line 4: COUNTRY
```

The following sample address illustrates the structure:

```
2101 MASSACHUSETTS AVE NW
WASHINGTON DC 20008-2811
USA
```

## Key Address Elements

Address Verification is unlikely to verify an address successfully if two or more of the following items are absent from the address:

- Province
- Locality
- ZIP Code

## Mapping Input Address Data to Address Elements

To verify an address, map the input address information to appropriate elements in Address Verification.

The following table shows how you can map the input address to the address elements:

| Information Type | Address Information | Address Elements |
|---|---|---|
| Street | MASSACHUSETTS AVE NW | Street Name, Street Directional |
| House Number | 2101 | HouseNumber |
| City | WASHINGTON | Locality |
| State | DC | ProvinceAbbreviation |
| ZIP Code | 20008-2811 | PostalCode |

## Input and Result XML in Address Verification (On-Premises)

The following example shows how you can map the address information to the XML address elements in Address Verification (On-Premises).

You enter the following input address:

```
<InputData>
     <AddressElements>
        <Country Item="1" Type="NAME">USA</Country>
        <Locality Item="1" Type="COMPLETE">WASHINGTON</Locality>
        <PostalCode Item="1" Type="UNFORMATTED">20008-2811</PostalCode>
        <Province Item="1" Type="COUNTRY_STANDARD">DC</Province>
        <Street Item="1" Type="COMPLETE">MASSACHUSETTS AVE NW</Street>
        <Number Item="1" Type="COMPLETE">2101</Number>
     </AddressElements>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result:

```
<?xml version="1.0" encoding="UTF-16"?>

<!-- EngineVersion="5.11.0.41264" -->

<Result     ProcessStatus="V4"
     ModeUsed="INTERACTIVE"
     Count="1"
     CountOverflow="NO"
     CountryISO3="USA"
     PreferredScript="LATIN_1"
     PreferredLanguage="ENGLISH">

<ResultData ResultNumber="1"
     MailabilityScore="5"
     ResultPercentage="100.00"
     ElementResultStatus="FFF0F8F0F000000000E0"
     ElementInputStatus="66606060600000000060"
     AddressResolutionCode="00000000000000000000"
     ExtElementStatus="00000000000000000000"
     ElementRelevance="11101010100000000010"
     AddressType="S"
     LanguageISO3="ENG">
    <AddressElements>
        <Country Type="NAME_EN" Item="1">United States</Country>
        <Locality Item="1">Washington</Locality>
        <PostalCode Item="1">20008-2811</PostalCode>
        <Province Item="1">DC</Province>
        <Province Item="2">District of Columbia</Province>
        <Street Item="1">Massachusetts Ave NW</Street>
        <Number Item="1">2101</Number>
    </AddressElements>
    <AddressLines>
        <DeliveryAddressLine Line="1">2101 Massachusetts Ave NW</DeliveryAddressLine>
        <CountrySpecificLocalityLine Line="1">Washington DC 20008-2811</
CountrySpecificLocalityLine>
        <FormattedAddressLine Line="1">2101 Massachusetts Ave NW</FormattedAddressLine>
        <FormattedAddressLine Line="2">Washington DC 20008-2811</FormattedAddressLine>
        <FormattedAddressLine Line="3">United States</FormattedAddressLine>
    </AddressLines>
    <AddressComplete>2101 Massachusetts Ave NW
 Washington DC 20008-2811
 United States</AddressComplete>
</ResultData>
</Result>
```

## Input and Result XML in Address Verification

The following example shows how you can verify a United States address in a SOAP call to Address Verification.

You submit the following request:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:v="http://
validator5.AddressDoctor.com/Webservice5/v4">
  <soap:Header/>
  <soap:Body>
    <v:Process>
        <v:login>*****</v:login>
        <v:password>******</v:password>
        <v:parameters>
           <v:ProcessMode>BATCH</v:ProcessMode>
           <v:ServiceParameters></v:ServiceParameters>
           <v:ValidationParameters>
           </v:ValidationParameters>
        </v:parameters>
        <v:addresses>
         <v:Address>
            <v:HouseNumber>
               <v:string>2101</v:string>
            </v:HouseNumber>
            <v:Street>
               <v:string>MASSACHUSETTS AVE NW</v:string>
            </v:Street>
            <v:PostalCode>
               <v:string>20008-2811</v:string>
            </v:PostalCode>
            <v:Locality>
               <v:string>WASHINGTON</v:string>
            </v:Locality>
            <v:Province>
               <v:string>DC</v:string>
            </v:Province>
            <v:Country>
               <v:string>USA</v:string>
            </v:Country>
         </v:Address>
        </v:addresses>
    </v:Process>
  </soap:Body>
</soap:Envelope>
```

Informatica Address Verification returns the following response:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
   <soap:Body>
      <ProcessResponse xmlns="http://validator5.AddressDoctor.com/Webservice5/v4">
         <ProcessResult>
            <StatusCode>100</StatusCode>
            <StatusMessage>OK</StatusMessage>
            <Results>
               <Result>
                  <ProcessStatus>V4</ProcessStatus>
                  <CountryISO3>USA</CountryISO3>
                  <ResultDataSet>
                     <ResultData>
                        <ResultNumber>1</ResultNumber>
                        <MailabilityScore>5</MailabilityScore>
                        <ResultPercentage>100.00</ResultPercentage>
                        <ElementInputStatus>6660606060000000060</ElementInputStatus>
                        <ElementResultStatus>FFF0F8F0F000000000E0</ElementResultStatus>
                        <ElementRelevance>11101010100000000010</ElementRelevance>
                        <ExtElementStatus>00000000000000000000</ExtElementStatus>
                        <AddressResolutionCode>000000000000000000000</AddressResolutionCode>
                        <AddressType>S</AddressType>
                        <LanguageISO3>ENG</LanguageISO3>
                        <Address>
                           <Street>
                              <string>MASSACHUSETTS AVE NW</string>
                           </Street>
                           <HouseNumber>
                              <string>2101</string>
```

```
                              </HouseNumber>
                              <Locality>
                                 <string>WASHINGTON</string>
                              </Locality>
                              <PostalCode>
                                 <string>20008-2811</string>
                              </PostalCode>
                              <Province>
                                 <string>DC</string>
                                 <string>DISTRICT OF COLUMBIA</string>
                              </Province>
                              <Country>
                                 <string>UNITED STATES</string>
                              </Country>
                              <DeliveryAddressLines>
                                 <string>2101 MASSACHUSETTS AVE NW</string>
                              </DeliveryAddressLines>
                              <CountrySpecificLocalityLine>
                                 <string>WASHINGTON DC 20008-2811</string>
                              </CountrySpecificLocalityLine>
                              <FormattedAddress>
                                 <string>2101 MASSACHUSETTS AVE NW</string>
                                 <string>WASHINGTON DC 20008-2811</string>
                                 <string>UNITED STATES</string>
                              </FormattedAddress>
                              <AddressComplete>2101 MASSACHUSETTS AVE NW;WASHINGTON DC
20008-2811;UNITED STATES</AddressComplete>
                           </Address>
                        </ResultData>
                     </ResultDataSet>
                  </Result>
               </Results>
            </ProcessResult>
         </ProcessResponse>
      </soap:Body>
   </soap:Envelope>
```

## Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified United States addresses.

The following table lists the types of enrichments that you can add to a United States address:

| Enrichment Type | Description |
|---|---|
| CAMEO | You can retrieve CAMEO profiles as enrichments for valid United States addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address. |
| Country-Specific Address Enrichments | You can retrieve the following enrichments for valid United States addresses:<br>- CBSA_ID. A five-digit number that identifies a Core-Based Statistical Area (CBSA). A CBSA is an urban area with a population greater than 10,000.<br>- CENSUS_BLOCK_GROUP. A 12-digit number that identifies a Census Block Group. A Census Block Group is a group of Census Blocks that share the same first digit.<br>- CENSUS_BLOCK_NO. A four-digit number that identifies a Census Block. A Census Block is the smallest entity for which the Census Bureau collects census information.<br>- CENSUS_TRACT_NO. A six-digit number that identifies a Census Tract. A Census Tract is a statistical subdivision of a county.<br>- CMSA_ID. A four-digit number that identifies a Consolidated Metropolitan Statistical Area (CMSA). A CMSA is a metropolitan area that is composed of other metropolitan areas with a population greater than 1,000,000.<br>- COUNTY_FIPS_CODE. A three-digit number that identifies a county within a state.<br>- FINANCE_NUMBER. A six-digit number, where the first two digits identify the state and the final four digits identify the post office or postal facility.<br>- MCD_ID. A five-digit number that identifies a Minor Civil Division (MCD). An MCD is a political or administrative area within a county.<br>- MSA_ID. A four-digit number that identifies a Metropolitan Statistical Area (MSA). An MSA is an urban area with a population greater than 50,000.<br>- PLACE_FIPS_CODE. A five-digit number that identifies a locality within a state.<br>- PMSA_ID. A four-digit number that identifies a Primary Metropolitan Statistical Area (PMSA). A PMSA is a metropolitan area that connects with one or more metropolitan areas to create an area with a population greater than 1,000,000.<br>- RECORD_TYPE. A single-character code that describes the type of mailbox, for example street or post office box.<br>- STATE_FIPS_CODE. A two-digit number that identifies a state in the United States.<br>- TIME_ZONE_CODE. A one- to three-character numeric value that indicates the offset between the address time zone and Greenwich Mean Time.<br>- TIME_ZONE_NAME. A three-character code that identifies the time zone to which the address belongs. |
| Standard Geocoding | You can retrieve the standard, interpolated geocoordinates for valid United States addresses. |
| Arrival Point Geocoding | You can retrieve the arrival point geocoordinates for valid United States addresses. |
| Parcel Centroid Geocoding | You can retrieve the parcel centroid geocoordinates for valid United States addresses. |

## Address Verification Tips and Best Practices

Note the following points when you verify United States addresses:

- The United States Postal Service recognizes a range of Spanish words and abbreviations.

The following table lists common Spanish address terms and the corresponding abbreviations:

| Spanish Words | Abbreviation |
|---|---|
| AVENIDA | AVE |
| CALLE | CLL |
| CAMINITO | CMT |
| CAMINO | CAM |
| CERRADA | CER |
| CIRCULO | CIR |
| ENTRADA | ENT |
| PASEO | PSO |
| PLACITA | PLA |
| RANCHO | RCH |
| VEREDA | VER |
| VISTA | VIS |

- If your input data contains sub-building data and street data in a single discrete field, Address Verification can parse the sub-building data from the street data.
- When you enter an address in a partially-fielded input format, limit the amount of non-address data in the delivery address line input.

  For example, the following input address includes non-address data:

```
<InputData>
    <AddressElements>
      <Country Item="1" Type="NAME">USA</Country>
      <Locality Item="1" Type="COMPLETE">FORT MYERS</Locality>
      <PostalCode Item="1" Type="UNFORMATTED">33916</PostalCode>
      <Province Item="1" Type="COUNTRY_STANDARD">FL</Province>
    </AddressElements>
    <AddressLines>
     <DeliveryAddressLine Line="1">VA OUTPATIENT CENTER-FT. MYERS</DeliveryAddressLine>
     <DeliveryAddressLine Line="2">3033 WINKLER EXTENSION</DeliveryAddressLine>
     <DeliveryAddressLine Line="3">LIC AV8635469</DeliveryAddressLine>
    </AddressLines>
</InputData>
```

  In the DeliveryAddressLine elements, the details such as `VA OUTPATIENT CENTER-FT. MYERS`, `WINKLER EXTENSION`, and `LIC AV8635469` indicate non-address data.
- Do not include locality, postal code, province, and country information in delivery address line input. For best results, enter the information in country specific locality line input or in discrete input fields.
- To verify an address to delivery point level, submit the address in certified mode. Address Verification can return Delivery Point Validation (DPV) fields for valid addresses in certified mode.
- To determine if a physical access point exists for an address, submit the address in certified mode and select the Door Not Accessible (DNA) element. A DNA address might not provide a door for mail delivery, or the mailbox might reside behind a locked gate.

- To verify or update an address to the form that the 911 emergency system recognizes, submit the address in certified mode.

  Use certified mode to convert rural route or highway addresses to street addresses.
- For best results, provide a descriptor when you enter sub-building information.

  For example, enter the sub-building information as `100 Main St Apt 10` instead of `100-10 Main St`, `100 Main St 10`, or `100 Main St #10`.
- For best results, use the ISO two-letter code or three-letter code to identify the country.
- To verify street alias names, Address Verification requires the `USA5C112.MD` reference data file. Address Verification requires the file in batch and interactive modes in addition to certified mode.

# Frequently Asked Questions

**How does Address Verification handle street information when two or more directional elements appears with the street name?**

Address Verification treats the first directional element as a pre-directional element and the second directional element as a part of the street name.

For example, in the street information `WEST NORTH ST`, the pre-directional element is `WEST` and the street name is `NORTH ST`.

**Can Address Verification tell whether an input address is a business or a home address?**

The Residential Data Indicator indicates whether the address points to a residence or a business.

**How does Address Verification return country information for overseas United States territories?**

Address Verification returns the country name as the United States for the following territories: Guam, Puerto Rico, and the Virgin Islands, U.S.

**Does Address Verification reference data include DPV data?**

Address Verification reference data for the United States contains DPV data. The certified DPV database has information that helps you check whether a ZIP+4 coded address is in the USPS delivery file as a known address record.

**How does Address Verification handle house numbers in different process modes?**

Batch and interactive modes can verify whether a house number is valid within the known range of numbers for a given street. However, batch and interactive modes cannot determine if the house number represents a deliverable address on the street. To verify that a house number identifies a deliverable address on a given street, verify the address in certified mode.

Certified mode can read additional address reference data files, such as LACSLink and SuiteLink files, and can perform additional calculations to confirm a unique delivery point for an address.

# Reference Address Databases

Informatica updates the United States reference address databases monthly. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the UpData newsletter from Informatica.

Informatica Address Verification provides the following reference databases for United States addresses:

- `USA5BI.MD`. Batch and interactive modes.
- `USA5FC.MD`. Fast completion mode.

- `USA5C101.MD` through `USA5C128.MD`. Certified mode.
- `USA5GC.MD`. Standard geocoding.
- `USA5GCAP.MD`. Arrival point geocoding.
- `USA5GCPC.MD`. Parcel centroid geocoding.
- `USA5E1.MD`. Supplementary data.
- `USA5CA.MD`. CAMEO data.

**Note:** You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification runs on Informatica Cloud.

## Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor-cloud

## Author

**Shahani Natalia Mendonca**

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